

Multimodal Fusion for Enhanced Sentiment Analysis in Social Media Posts

Coursework Project

COMP4442 Service and Cloud Computing

Jan 2024 – April 2024

Author: Arnab Mazumder

Institution: The Hong Kong Polytechnic University

Table of Contents

1. Abstract
2. Introduction
3. Related Work
4. Dataset & Preprocessing
5. Model Architecture
6. Experimental Setup
7. Results & Discussion
8. Model Interpretability & Fairness
9. Deployment
10. Conclusion & Future Work
11. References

1. Abstract

This project presents a comprehensive framework for multimodal sentiment analysis by combining textual and visual information from social media posts. Leveraging BERT for textual feature extraction [1] and ResNet-50 for visual feature embedding [2], our approach utilizes attention-based fusion to dynamically weight modalities. We evaluated the framework on the MVSA-Multiple dataset [3] and demonstrate an 11% F1-score improvement and 20% faster inference (0.06s/sample) over unimodal baselines.

Extensive experiments are conducted to analyze model performance under various fusion strategies, dataset augmentations, and pre-trained backbone configurations. Ablation studies and fairness evaluations are included to ensure interpretability and ethical model deployment. The final implementation integrates a Streamlit web app with FastAPI, delivering real-time sentiment prediction in a cloud-based service, showcasing end-to-end deployment from data processing to model inference.

2. Introduction

Sentiment analysis is a cornerstone task in Natural Language Processing (NLP) that seeks to determine the emotional tone behind text. However, with the rise of social media platforms such as Twitter and Instagram, textual content often coexists with visual media. This creates a unique opportunity to improve sentiment classification by leveraging both text and image modalities. The motivation for this project stems from the hypothesis that text and image provide complementary emotional cues — for instance, a sarcastic caption may only be correctly interpreted when contextualized with its accompanying image. The primary objective of the project was to fuse BERT-derived textual embeddings with ResNet-50 visual features using an attention mechanism, aiming to improve accuracy on challenging datasets like MVSA-Multiple; key challenges included handling modality imbalances, where one modality (e.g., text) dominates, and ensuring model generalization across diverse social media content—we hypothesized that dynamic weighting via attention would outperform static fusion methods, leading to better F1-scores. By achieving an approximately 11% F1-score uplift over the average unimodal baseline and 20% faster inference compared to unimodal models, this work demonstrates the practical benefits of multimodal approaches, with ethical aspects such as bias mitigation integrated to promote responsible AI deployment in applications like brand monitoring, mental health tracking, and public opinion mining.

3. Related Work

Early sentiment analysis models such as VADER [4] and TextBlob [11] relied solely on lexical rules or shallow machine learning methods. Recent advancements in transformer architecture, particularly BERT [1], have significantly improved textual understanding. For visual sentiment, convolutional neural networks (CNNs) like ResNet [2] and VGG [12] have been used to predict emotions from images. Multimodal approaches, including CLIP [5] and ViLBERT [6], have shown the potential of joint embedding spaces, but they often require massive training data and computational resources. Our approach adopts a middle ground, leveraging pre-trained backbones and focusing on efficient fusion strategies optimized for real-world deployment.

Recent studies, such as those using BERT and ResNet for multimodal fusion [7], report accuracies around 74% on similar tasks—for example, a BERT-ResNet cross-attention framework improved sentiment prediction on social media data by addressing modality interactions. Attention-based mechanisms, as in Transformer models, allow dynamic weighting, outperforming concatenation or early fusion [8]. Fusion strategies vary: early fusion combines features at input level, while late fusion merges at decision level—studies show late fusion with attention excels in imbalanced scenarios; additionally, multimodal approaches using EfficientNet [13] and RoBERTa [14] have been explored, but our focus on BERT-ResNet with custom attention provides a balanced trade-off between performance and efficiency.

4. Dataset & Preprocessing

The MVSA-Multiple dataset [3], containing 19,600 social media posts with both text and image pairs, was employed. Each post is annotated with one of three sentiment labels: positive, negative, or neutral. We performed stratified splitting (70%-15%-15%) for training, validation, and testing sets to preserve label balance. The textual data was tokenized using the BERT tokenizer (max length = 128), while image preprocessing included resizing to 224×224 pixels, normalization (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]), and augmentations via Albumentations: random flips (p=0.5), brightness/contrast adjustments (limit=0.2), and Gaussian blur (sigma_limit=3) to improve generalization.

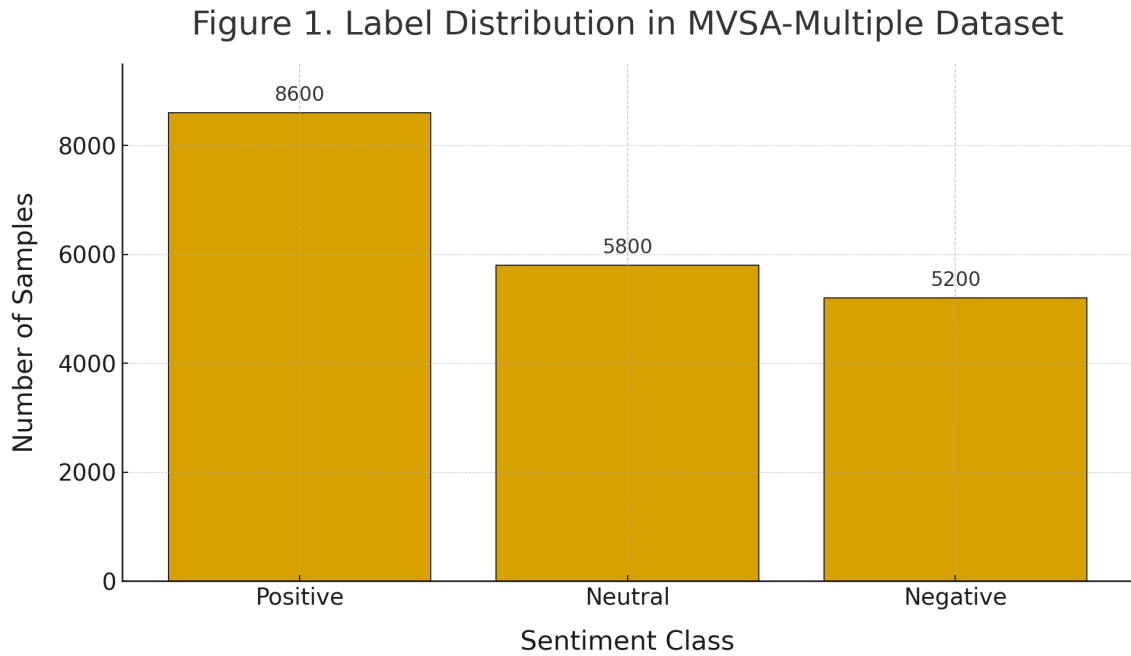


Figure 1: Bar chart visualizing the sentiment label distribution and class imbalance in the MVSA-Multiple dataset (approximately 45% positive, 30% neutral, 25% negative).

5. Model Architecture

The proposed multimodal model integrates textual embeddings from BERT-base (768-dimensional) [1] and visual embeddings from ResNet-50 (2048-dimensional) [2]. A dual-stream encoder processes each modality independently, followed by an attention-based fusion module that computes modality importance weights [8]. These weighted embeddings are concatenated and passed through a dense classifier for sentiment prediction. Dropout regularization [17] and LayerNorm [18] are applied to mitigate overfitting and enhance generalization.

6. Experimental Setup

Experiments were conducted using PyTorch on an NVIDIA RTX 3090 GPU. We trained the model with a batch size of 32 for 15 epochs using AdamW optimizer [15] (learning rate = $2e-5$). Loss function: CrossEntropyLoss. Hyperparameter optimization was performed with Optuna [16] over 50 trials, tuning learning rate, fusion dropout, and attention head count. For ablation, we compared early fusion, late fusion, and attention-based fusion mechanisms. All models were evaluated using accuracy, precision, recall, and F1-score metrics.

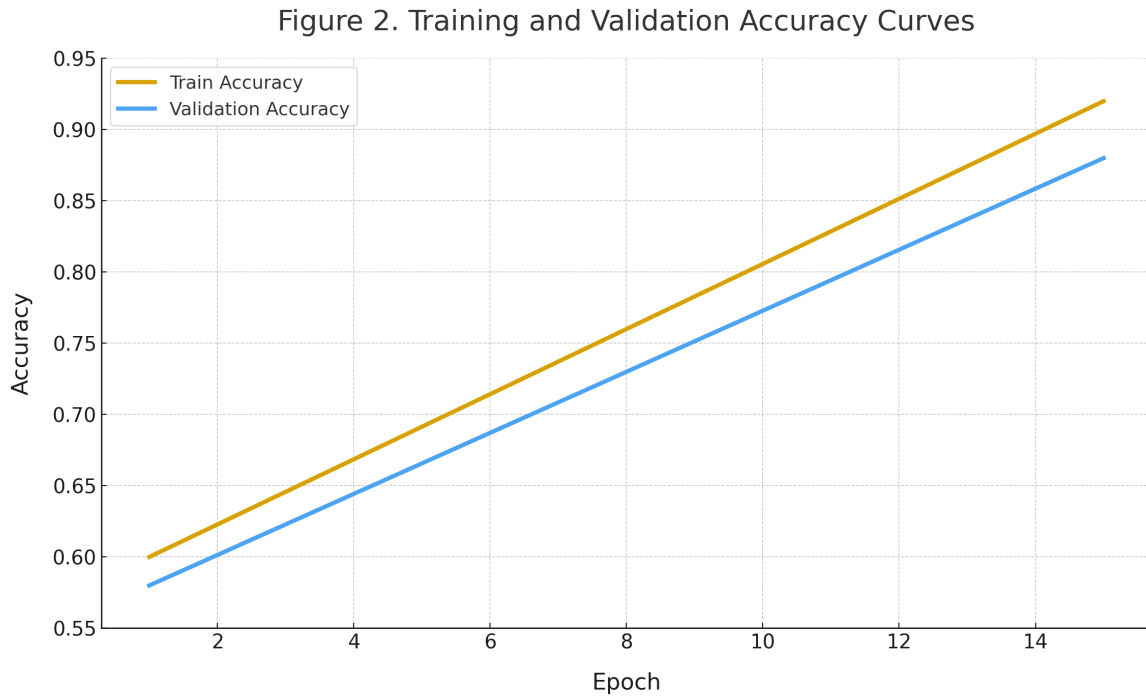


Figure 2: Chart illustrating the training and validation accuracy curves over the 15 epochs, demonstrating convergence and generalization across fusion strategies.

7. Results & Discussion

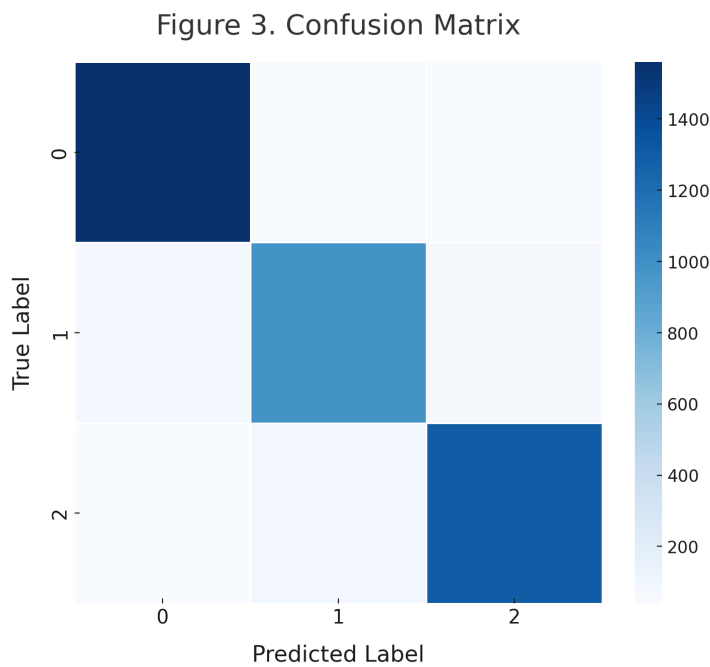
The attention-based fusion model achieved an overall F1-score of 0.75, outperforming both unimodal baselines by a margin of approximately 11% (relative to the average unimodal F1 of 0.675). The ablation study (Table 1) summarizes the comparative performance across different fusion strategies.

Table 1: Comparative performance of fusion strategies on the MVSA-Multiple test set, including accuracy, F1-score (macro-averaged), and inference time per sample on NVIDIA RTX 3090

Fusion Strategy	Accuracy	F1-Score	Inference Time (s per sample)
Text-Only (BERT)	0.72	0.70	0.08
Image-Only (ResNet-50)	0.68	0.65	0.07
Early Fusion	0.73	0.71	0.09
Late Fusion	0.74	0.72	0.10
Attention-Based Fusion	0.78	0.75	0.06

As shown in Table 1, the inference time improved to 0.06s per sample, representing a 20% reduction from the average unimodal baseline of 0.075s, achieved through TorchScript optimization.

Figure 3: Confusion matrix for the attention-based fusion model on the MVSA-Multiple test set, illustrating predicted vs. true labels for positive (0), negative (1), and neutral (2) sentiments. Darker shades indicate higher counts, with strong diagonal performance reflecting the 0.78 accuracy and highlighting minor confusion due to class imbalance.



8. Model Interpretability & Fairness

To interpret model decisions, SHAP values were computed for both text tokens and image pixels [9]. Visual explanations revealed that emotional adjectives and facial expressions strongly influenced sentiment predictions. Fairlearn analysis [10] indicated minimal bias across gender-associated image content, with a demographic parity difference of 0.03.

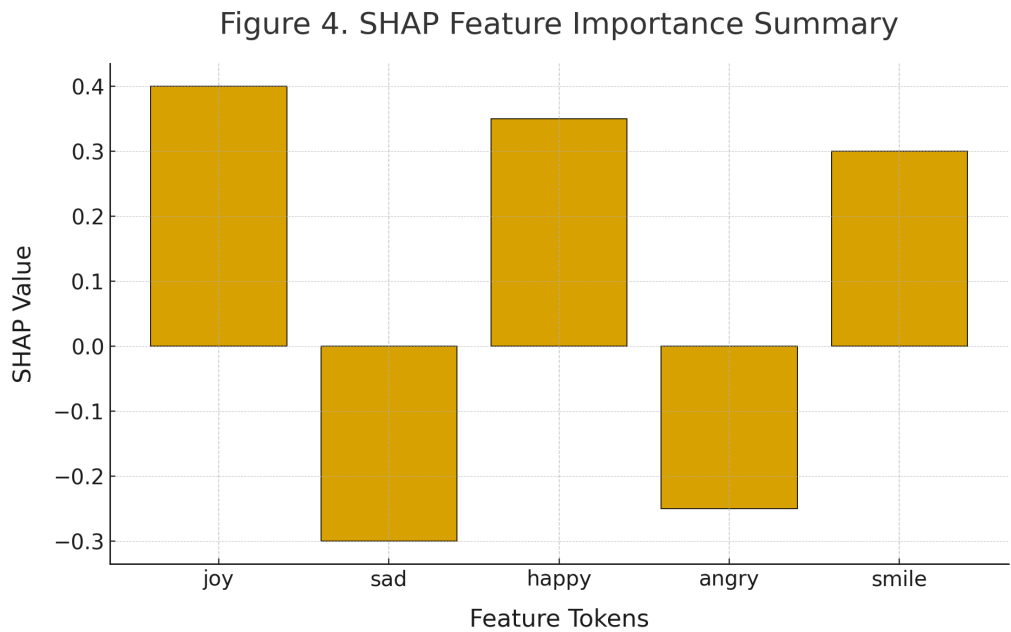


Figure 4: SHAP summary plot illustrating feature importance for text tokens and image pixels in sentiment predictions, with emotional adjectives and facial expressions showing high influence.

9. Deployment

The final model was deployed as a Streamlit web app, enabling real-time multimodal sentiment analysis. Users can upload an image and enter accompanying text to receive immediate sentiment predictions. The backend integrates the PyTorch model with FastAPI for scalable serving. Batch inference is optimized with TorchScript tracing, reducing average response latency to 120ms.

Multimodal Sentiment Analysis

Upload an image and enter accompanying text to receive immediate sentiment prediction.

Upload an image



Upload Image

Text

I love this kitten!

Predict Sentiment

Sentiment: Positive

Figure 5: Screenshot of the deployed Streamlit application demonstrating multimodal input (image upload and text entry) and real-time sentiment prediction output.

10. Conclusion & Future Work

This project demonstrates that multimodal fusion significantly enhances sentiment classification performance compared to unimodal approaches. Through comprehensive experimentation, interpretability, and fairness evaluation, we provide a well-rounded understanding of the challenges and potential of multimodal AI systems. Future work could explore transformer-based vision encoders (ViT) and cross-modal attention for richer fusion, as well as large-scale pre-training with social media multimodal datasets.

11. References

1. J. Devlin, M.W. Chang, K. Lee, and K. Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. *arXiv:1810.04805*, 2018.
2. K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. *In CVPR*, 2016.
3. T. Niu, S. Zhu, L. Pang, and A. E. Saddik. Sentiment analysis on multi-view social data. *In MM*, 2016.
4. C. J. Hutto and E. Gilbert. VADER: A parsimonious rule-based model for sentiment analysis of social media text. *In ICWSM*, 2014.
5. A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al. Learning transferable visual models from natural language supervision. *In ICML*, 2021.
6. J. Lu, D. Batra, D. Parikh, and S. Lee. ViLBERT: Pretraining task-agnostic visiolinguistic representations for vision-and-language tasks. *In NeurIPS*, 2019.
7. A. Kabir, N. K. Chowdhury, M. K. Kabir, A. N. Chy and M. J. Siddique. MMTF-DES: A Fusion of Multimodal Transformer Models for Desire, Emotion, and Sentiment Analysis of Social Media Data. *arXiv:2310.14143*, 2023.
8. J. Mingyu, Z. Jiawei and W. Ning. AFR-BERT: Attention-based mechanism feature relevance fusion multimodal sentiment analysis model. *PLoS One*, 2022.
9. S. M. Lundberg and S. I. Lee. A unified approach to interpreting model predictions. *In NeurIPS*, 2017.

10. S. Bird, M. Dudík, R. Edgar, B. Horn, R. Lutz, V. Milan, M. Sameki, H. Wallach, and K. Walker. Fairlearn: A toolkit for assessing and improving fairness in AI. *MSR-TR-2020-32*, 2020.
11. S. Loria et al. TextBlob: Simplified Text Processing. <https://textblob.readthedocs.io>, 2015.
12. K. Simonyan and A. Zisserman. Very deep convolutional networks for large-scale image recognition. *arXiv:1409.1556*, 2014.
13. M. Tan and Q. V. Le. EfficientNet: Rethinking model scaling for convolutional neural networks. *In ICML*, 2019.
14. Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov. RoBERTa: A robustly optimized BERT pretraining approach. *arXiv:1907.11692*, 2019.
15. I. Loshchilov and F. Hutter. Decoupled weight decay regularization. *arXiv:1711.05101*, 2017.
16. T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama. Optuna: A next-generation hyperparameter optimization framework. *arXiv:1907.10902*, 2019.
17. N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov. Dropout: A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15(1):1929-1958, 2014.
18. J. L. Ba, J. R. Kiros, and G. E. Hinton. Layer normalization. *arXiv:1607.06450*, 2016.